1.6 The ε-δ Definition of a Limit

Name _____ Period ____ Date _____

AP Calculus 1.6 · Student sheet

I can…

- explain what ε and δ measure.
- find the largest δ that works for a given ε.
- write an ε-δ proof for the limit of a linear function.

Vocabulary · fill in the missing word

ε-δ definition of a limit — the limit of f(x) as x → c is L if for every ε > 0 there is a δ > 0 so that 0 < |x − c| < δ _____ |f(x) − L| < ε

Part 1 · Target practice

f(x) = 2x near x = 1, where the limit is 2.

ε	0.5	0.1	0.01
Largest δ			

Part 2 · Notes and examples (fill in each note, then work the example)

What they measure: ε is how close f(x) must be to _____; δ is how close x must be to c.

Example 1

The limit of 2x as x → 1 is 2. Find the largest δ that works for ε = 0.01.

Definition: for every ε > 0 there is δ > 0 so that 0 < |x − c| < δ implies |f(x) − L| < _____.

Example 2

The limit of −3x as x → 2 is −6. Find the largest δ for ε = 0.01.

Linear functions: for f(x) = mx + b, m ≠ 0, δ = _____ works.

Example 3

f(x) = 4x − 1 and the limit as x → 3 is 11. Find δ for ε = 0.1, for ε = 0.01, and for any ε > 0.

Writing the proof: “Given ε > 0, let δ = …” and end by showing |f(x) − L| _____ ε.

Example 4

Prove that the limit of $2x + 5$ as $x \to 0$ is 5.

Example 5

Prove that the limit of x^2 as $x \to 2$ is 4.

Part 3 · Partner practice

Partner 1

The limit of $2 - 5x$ as $x \to -2$ is 12. Find the largest δ for $\varepsilon = 0.02$.

Partner 2

Prove that the limit of $3x$ as $x \to 3$ is 9.